

BME 3113 “Wearable Technology Studio” Syllabus (Fall 2026)

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Studio/Office hours: TBD

Class hours: Section 01 Tuesday, 2:00 – 4:50 pm

Course Content

This course will focus on product design, prototyping, entrepreneurship and the innovation process for wearable technology by emphasizing creation of solutions through team-based projects. Student teams will utilize tools and skills such as computer-aided design, fabrication methods and sensor feedback to create prototypes of various types of wearable devices. Students will thus learn about the working principles, design, manufacture, and usability of wearable technology. They will investigate market interest in an open-ended wearable concept and communicate their solution's societal and economic benefits as well as technical, regulatory and manufacturing issues related to product design.

Course Instructional Format

The fundamental objective of this course is to allow students to learn by doing. The class will explore of various tools and resources for engineering design, participate in-class skill building activities, and complete group projects. Students will be assessed based on digital portfolios of individual work, group project notebooks, website documentations and presentations, as well as in-class and group participation.

Skill Building Activities

- Review online topical reference materials and tutorials that are provided each week on Canvas.
- Individually complete the skill demonstration activity.
- Organize an individual portfolio documenting each of the skill building processes.

Group Projects

- Groups will be assigned for each project.
- Weekly updates with specified project goals and logistics.
- Groups will complete the assigned project deliverables outside of class times. Attendance at “Open Studio Hours” or a Zoom meeting with faculty during “Office Hours” is expected on a regular basis to complete fabrication and troubleshooting. These meetings should be scheduled in advance.
- “Pin-Up Review” presentations will facilitate peer feedback on initial ideas and designs.
- Group project notebooks documenting weekly progress will be submitted for grading.
- Projects will conclude with documentation of specified information as well as a group presentation and demonstration of the final design. Specific deliverables and documentation details will be provided for each project.

Materials and Resources

The hands-on nature of this course will require different materials for prototyping. Many of these materials will be provided by instructors (including textile materials, conductive thread/fabric, sewing needles, sensors, microprocessor boards, lithium batteries, adapters, LEDs, pedometers, etc.). Some materials may need to be sourced individually or as a team (including generic fabric, LED strips, foam, elastic bands, microUSB cords, and generic tools such as screwdrivers). All tools and materials in the lab are available for student use in class and during open studio hours.

Learning objectives

- Identify needs and opportunities as well as the process of innovation in Wearable Technology.
- Identify constraints, human factors and best practices in the design of Wearable Technology.
- Collaborate in teams with defined roles and responsibilities and common goals to organize, plan and manage a product design project.
- Utilize technical tools and skills needed to create a viable design solution in terms of economic and social value.
- Report on the quality control, regulation, privacy and safety aspects of wearables.
- Communicate design status and results to all stakeholders in verbal, written, and public presentation formats at appropriate points in the development timeline.
- Recognize the contribution of science, technology, engineering and/or mathematics to society

Grading/Evaluation:

- **10% Attendance and Group Participation**

- It is important that students attend the class on time. Missing days will hurt your grade in this course. Please contact the instructor for any and all anticipated absences, and especially emergencies relating to class attendance. The student will get an “F” after the second unexcused absence.

Participation Grade	Perfect Attendance	1 Absence	2 Absences
A	10%	8%	Fail
B	8%	6%	Fail
C	7%	5%	Fail
D	6%	4%	Fail
F	5%	3%	Fail

A = Fully engaged in class discussions. Asked and answered questions from instructor/class.

B = Mostly engaged in class and team discussions. Occasionally asked and answered questions from instructor and classmates. Occasionally distracted or engaged in other activities.

C = Only partially engaged in class discussions. Only responded when directly asked a question by the instructor. Occasionally distracted or engaged in other activities.

D = Rarely participated in class discussions. Only partially able to answer questions directly posed by professor. Frequently distracted or engaged in other activities.

F = Attended class but did not actively participate. Unable to answer directly posed questions.

- **30% Individual Coursework**

- Skill building activities will require students to follow tutorials and complete a demonstration prototype. The activity process will be documented in a Skills Portfolio.
- Online materials and software will be used to create interactive presentation slides about specific topics that can be shared with the other students.

- **20% Group Project Updates**

- Each week there will be assigned project deliverables related to the topics discussed. Project documentation in the form of a group notebook will be submitted to Canvas.
- Preliminary project ideas will be discussed with other students through “Pin-up Review” presentations. Peer feedback will be used to update the project designs.

- **40% Project Presentations**

- Three group projects will conclude with documentation of specified deliverables and an in-class final design demonstration presentation.
- Self and group participation will be assessed by students and faculty.

Grading Scale:

A	> 93%
A-	90-92.9%
B+	87-89.9%
B	83-86.9%
B-	80-82.9%
C+	77-79.9%
C	73-76.9%
C-	70-72.9%
D+	67-69.9%
D	60-66.9%
F	<60%

Course Conduct and Academic Integrity:

Students should read and comply with the Lawrence Technological University Student Code of Conduct: <https://ltu.edu/student-life/student-dean/code-of-conduct/>

Academic honesty and integrity are fundamental values of the University. Cheating will not be tolerated. This includes plagiarism or copying assignments from any unauthorized source. Any material from an authorized source should be properly cited. Students may discuss homework problems, but should write up solutions independently (or as part of their assigned team). Violations will be reported as Academic Misconduct in accordance with the procedures outlined in the Student Code of Conduct.

AI tools may be used for brainstorming, research assistance, coding support, or feedback on drafts of your work. For some assignments, AI use may be encouraged or even necessary to complete an assignment. However, submitting fully AI-generated artifacts as your own work (without explicit permission and clear disclosure) counts as plagiarism and unauthorized aid. Submitted work may be screened using plagiarism and AI-detection tools. If your work is flagged, your instructor may ask to discuss it with you. If you cannot demonstrate understanding of what you, or your team members, submitted by recalling, explaining, and defending the work (or providing other supporting materials), you may receive a zero for that assignment and be reported for Academic Misconduct in accordance with the procedures outlined in the Student Code of Conduct.

You may be required to write the following pledge on academic work submitted along with your signature:

“I have neither given nor received unauthorized aid in completing this work, nor have I presented someone else’s work as my own.”

Semester Schedule (tentative)

Week	Dates	Topics	Individual Assignments	Group Project	Project Deliverables
1	8/25	What is Wearable Tech?	Teardown Wearable Profile	Project 1: “Light-up Costume”	Ideation, Design Sketch
2	9/1	Textiles	Hand Stitching Sewing Machine	Textile Design	Textile Prototype GitHub Page
3	9/8	Circuits and Electronics	TinkerCAD Circuits + Blue Devil Boards	LED/Battery Design	LED Prototype Logistics Checklist
4	9/15	3D Design/ Printing	TinkerCAD Design	“Case” Design	Instructable Story Template
5	9/22	Microcontrollers + Sensors/Actuators	Adafruit Bluefruit	Project 2: “Mixed Reality Tracker”	P1 Presentation
6	9/29	AI Coding Agents	CircuitPython	Programming Prototype	Pinup: Design Concept / Code
7	10/6	User Interaction / HCD	Accessibility Design	Usability/UX Design	GN: UX Design
8	10/13	Industrial Design and Ergonomics	Ergonomics	“Fit and Finish” Design	Pinup: Industrial Design
9	10/20	Tech Revolutions	Extended Reality ADL Tracker	Project 3: “Wearable Assistant”	P2 Presentation
10	10/27	Machine Learning	Computer Vision	Algorithm Design	Algorithm Design
11	11/3	Big Data	Data Visualization	Data Engineering	Data Display
12	11/10	Software	Figma	App Design	App Design
13	11/17	Privacy and Security	Fair Information Practices	Sustainability	Privacy Policy
14	11/24	Product Development	Business Model	Commercialization	“Kickstarter” page
15	12/1	Entrepreneurship	Career Reflection	Pitches	Marketing Pitch
FINAL	12/10	1:30PM – 3:30PM			P3 Presentation